LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: The Fundamental Unit of Life (Cell)

Milestones

- Robert Hooke (1665) saw cells in a thin slice of cork. Leeuwenhoek first saw free-living cells in pond water. Robert Brown discovered the nucleus (1831).
- **Cell theory:** all plants and animals are made of cells, and all cells arise from pre-existing cells.

Prokaryotic cell	Eukaryotic cell
Small; nuclear region (nucleoid) with no nuclear membrane	Larger; nucleus with a nuclear membrane
Single chromosome; no membrane-bound organelles	More than one chromosome; membrane-bound organelles present
e.g. bacteria	e.g. plant and animal cells

Cell organelles

Part	Job
Plasma membrane	Selectively permeable; lets only some substances in and out
Cell wall (plants)	Made of cellulose; gives strength and lets the cell survive in dilute solutions
Nucleus	Contains chromosomes (DNA); controls the cell's activities
Endoplasmic reticulum	Rough ER (with ribosomes) makes proteins; smooth ER makes fats; transport channels; detoxification
Golgi apparatus	Stores, modifies and packages products; forms lysosomes
Lysosomes	Digest foreign material and worn-out parts; 'suicide bags' of the cell
Mitochondria	Release energy as ATP; 'powerhouse'; have their own DNA and ribosomes
Plastids (plants)	Chloroplasts (photosynthesis); leucoplasts (store starch, oils, proteins)
Vacuoles	Store materials; very large in plant cells, small in animal cells

Osmosis: water moves through a selectively permeable membrane from a dilute solution to a concentrated one.
Cell in a **hypotonic** (dilute) solution swells · **isotonic**: no change · **hypertonic** (concentrated): shrinks.
Plasmolysis: a plant cell's contents shrink away from the wall when it loses water.

Quick check

1. Why are lysosomes called the 'suicide bags' of the cell?

2. What happens to a raisin placed in plain water, and why?